Dynamical Systems Homework #4

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Problem (Exercise 1). *Let $(X, \mathcal{B}, \mu, T)$ be a totally ergodic measure preserving system. Show that for each $f \in L^2(X, \mu)$, we have*

$$\frac{1}{N}\sum_{n=0}^{N-1} U_T^{n^2} f \xrightarrow[N\to\infty]{L^2} \int_X f d\mu.$$

Solution. We will use [Ber87, Theorem 1.4]. We restate it here.

Theorem 0.1 (Van der Corput's Difference Theorem). *If $\mathcal{H}$ is a Hilbert space and $(g_n)_{n\geq 0} \subseteq \mathcal{H}$ is a bounded sequence satisfying*

$$\lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} \langle g_{n+h}, g_n\rangle = 0,$$

for every $h \in \mathbb{N}$, then $\lim_{N\to\infty} \|\frac{1}{N}\sum_{n=0}^{N-1} g_n\| = 0$.

We will apply this theorem for $\mathcal{H} = L^2(X, \mu)$ and $g_n := U_T^{n^2} f$. Also without loss of generality assume $\int_X f d\mu = 0$ (otherwise we can always write $f = \alpha + f^*$ with $\alpha = \int_X f d\mu$ and $f^* = f - \alpha$). Now, let us evaluate the scalar product:

$$\langle g_n, g_{n+h}\rangle = \langle U_T^{(n+h)^2} f, U_T^{n^2} f\rangle = \langle U_T^{2nh} U_T^{h^2} f, f\rangle,$$

where in the last equality we used the fact that U_T is an isometry, *i.e.* $\langle U_T f, U_T g\rangle = \langle f, g\rangle$. Now we evaluate the limit of the average:

$$\lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} \langle g_n, g_{n+h}\rangle = \lim_{N\to\infty} \left\langle \frac{1}{N}\sum_{n=0}^{N-1} U_{T^{2h}}^n U_T^{h^2} f, f\right\rangle.$$

Here, we used the fact that $U_T^{2nh} = U_{T^{2h}}^n$.

Now we invoke the L^2 Mean Ergodic Theorem applied to the transformation T^{2h}:

$$\frac{1}{N}\sum_{n=0}^{N-1} U_{T^{2h}}^n (U_T^{h^2} f) \xrightarrow[N\to\infty]{L^2} P_{\mathcal{I}}(U_T^{h^2} f),$$

where $P_{\mathcal{I}} : L^2(X, \mu) \to \mathcal{I}$ is a projection and $\mathcal{I} = \{f \in L^2(X, \mu) : U_{T^{2h}} f = f\}$. Finally, we use the fact that $(X, \mathcal{B}, \mu, T)$ is *totaly ergodic*, implying T^{2h} is ergodic. Therefore, $P_{\mathcal{I}}(U_T^{h^2} g) = \int_X U_T^{h^2} f d\mu = \int_X f d\mu$ (last equality is due to T being measure-preserving). Since we assumed $\int_X f d\mu = 0$, we conclude that our chosen sequence $(g_n)_{n\geq 0}$ satisfies conditions of the theorem. Consequently,

$$\frac{1}{N}\sum_{n=0}^{N-1} U_T^{n^2} f \xrightarrow[N\to\infty]{L^2} \int_X f d\mu.$$

Problem (Exercise 2). *Let $k \in \mathbb{N}$, $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and let $T : \mathbb{R}^d/\mathbb{Z}^d \to \mathbb{R}^d/\mathbb{Z}^d$ be given by*

$$T : (x_1, x_2, \ldots, x_d) \mapsto (x_1 + \alpha, x_2 + x_1, x_3 + x_2, \ldots, x_d + x_{d-1}).$$

Show that the topological dynamical system $(\mathbb{R}^d/\mathbb{Z}^d, T)$ is uniquely ergodic.

Solution. As the hint suggests, we use Furstenberg's Theorem and prove the statement by induction. The base of induction is $(\mathbb{R}/\mathbb{Z}, T : x_1 \mapsto x_1 + \alpha)$ which is well-known to be both ergodic and uniquely ergodic. Now assume that $(\mathbb{R}^{d-1}/\mathbb{Z}^{d-1}, T_{d-1})$ with T_{d-1} given above is uniquely ergodic with invariant Lebesgue measure μ_{d-1}. Then, we can view our map T_d on $\mathbb{R}^d/\mathbb{Z}^d$ as a skew product over T_{d-1}: that is, we have $T_d(\mathbf{x}, x_d) = (T_{d-1}(\mathbf{x}), x_d + \pi(\mathbf{x}))$ where $\mathbf{x} = (x_1, \ldots, x_{d-1}) \in \mathbb{R}^{d-1}/\mathbb{Z}^{d-1}$ and $\pi : (x_1, \ldots, x_{d-1}) \mapsto x_{d-1}$ is a continuous projection. By Furstenberg's Theorem, to show T_d is uniquely ergodic, it is now sufficient to show that T_d is ergodic.

Ergodicity of T_d. Here we use Fourier-analytic approach. Let $F \in L^2(\mathbb{R}^d/\mathbb{Z}^d)$ and $F \circ T_d = F$. Expand F into Fourier series:

$$F(\mathbf{x}, x_d) = \sum_{m \in \mathbb{Z}} f_m(\mathbf{x}) e^{2\pi i m x_d} \text{ where } f_m \in L^2(\mathbb{R}^{d-1}/\mathbb{Z}^{d-1}) \text{ and } \sum_{m \in \mathbb{Z}} \|f_m\|_2^2 < \infty.$$

Now substitute into $F \circ T_d = F$:

$$\sum_{m \in \mathbb{Z}} f_m(T_{d-1}\mathbf{x}) e^{2\pi i m (x_d + x_{d-1})} = \sum_{m \in \mathbb{Z}} f_m(\mathbf{x}) e^{2\pi i m x_d},$$

implying $f_m(T_{d-1}\mathbf{x}) e^{2\pi i m x_{d-1}} = f_m(\mathbf{x})$ for all $m \in \mathbb{Z}$. For $m = 0$, this reduces to $f_0 \circ T_{d-1} = f_0$, implying f_0 is a constant function. Otherwise, write $f_m(T_{d-1}\mathbf{x}) = e^{-2\pi i m x_{d-1}} f_m(\mathbf{x})$. Expand f_m into Fourier series again:

$$f_m(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^{d-1}} \widehat{f}_{m,\mathbf{n}} e^{2\pi i \langle \mathbf{n}, \mathbf{x} \rangle}.$$

On the left-hand side we obtain:

$$f_m \circ T_{d-1}(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^{d-1}} \widehat{f}_{m,\mathbf{n}} e^{2\pi i E(\mathbf{x},\mathbf{n})}, \; E(\mathbf{x}, \mathbf{n}) = n_1 \alpha + \sum_{j=1}^{d-2} (n_j + n_{j+1}) x_j + n_{d-1} x_{d-1}.$$

On the right-hand side:

$$e^{-2\pi i m x_{d-1}} f_m(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^{d-1}} \widehat{f}_{m,\mathbf{n}} e^{2\pi i G(\mathbf{x},\mathbf{n})}, \; G(\mathbf{x}, \mathbf{n}) = \sum_{j=1}^{d-2} n_j x_j + (n_{d-1} - m) x_{d-1}.$$

Since $f_m \circ T_{d-1} = e^{-2\pi i m x_{d-1}} f_m$, coefficients of both expressions must match frequency by frequency. Inspect the exponent for x_{d-1}: on the right-hand side, the frequency is $n_{d-1} - m$, while on the left-hand side, it is n_{d-1}. Thus:

$$|\widehat{f}_{m,n_1,\ldots,n_{d-1}}| = |\widehat{f}_{m,n_1',\ldots,n_{d-1}+m}| = |\widehat{f}_{m,n_1'',\ldots,n_{d-1}+2m}| = \ldots,$$

forcing too many coefficients to be non-zero if $\widehat{f_m} \neq 0$. Thus, all coefficients are strictly 0 and thus $\widehat{f_m} = 0$. This concludes the proof that F is a constant and thus T_{d-1} is ergodic.

Problem (Exercise 3). *Let $p(n) = a_0 n^d + a_1 n^{d-1} + \cdots + a_{d-1} n + a_d$ be a polynomial sequence with real coefficients. Show that the following conditions are equivalent:*

(a) The coefficient a_i is irrational for at least one $0 \leq i < d$;

(b) The sequence $(p(n) \bmod 1)_{n \geq 0}$ is equidistributed (in $\mathbb{R}/\mathbb{Z}$, with respect to the Lebesgue measure).

Solution.

$\underline{(b) \Rightarrow (a)}$. We prove contrapositively (*i.e.*, $\neg(a) \Rightarrow \neg(b)$). Assume $a_i = p_i/q_i$ for $p_i \in \mathbb{Z}$, $q_i \in \mathbb{N}$, and let $q := \operatorname{lcm}(q_0, \ldots, q_{d-1})$.

Claim. $(p(n))_{n \geq 0}$ is periodic in $\mathbb{R}/\mathbb{Z}$ with a period at most q.

Proof. Compute $p(n+q) - p(n)$:

$$p(n+q) - p(n) = \sum_{k=0}^{d-1} a_{d-k}((n+q)^k - n^k).$$

Note that $(n+q)^k - n^k$ is always a multiple of q: say, $(n+q)^k - n^k := c_k q$. Then,

$$p(n+q) - p(n) = \sum_{k=0}^{d-1} p_{d-k} c_k \cdot \frac{q}{q_{d-k}} \in \mathbb{Z}.$$

Thus $p(n+q) = p(n)$ in $\mathbb{R}/\mathbb{Z}$.

$\underline{(a) \Rightarrow (b)}$.

Claim. Without loss of generality, we might assume $a_0 \in \mathbb{R} \setminus \mathbb{Q}$.

Proof. Assume k is the highest degree whose coefficient is irrational. Let $p_{\leq k}(n) = a_d + a_{d-1} n + \cdots + a_{d-k} n^k$ and $p_{>k}(n) = a_{d-k-1} n^{k+1} + \cdots + a_0 n^d$. Let q be the common denominator of rational coefficients in $p_{>k}(n)$. Also note that $(p_{>k}(n))_{n \geq 0}$ is periodic (see argument above). Let $n = qm + r$ for $0 \leq r < q$. Then, $p(qm+r) = p_{>k}(r) + p_{\leq k}(qm+r)$ in $\mathbb{R}/\mathbb{Z}$. Note that $p_{>k}(r)$ is just a constant while $p_{\leq k}(qm+r)$ is a polynomial of m of degree k with leading coefficient $a_{d-k} q^k$, which is irrational. If we prove the result of equidistributivity of $p_{\leq k}(qm+r)$, we are done.

Assuming $a_0 \in \mathbb{R} \setminus \mathbb{Q}$, we invoke Corollary 4.32 and so the statement holds.

Problem (Exercise 4). *Construct a sequence of ergodic measures for the full binary shift $\mu_n \in \mathcal{M}_e^\sigma(\{0,1\}^{\mathbb{N}})$ that converges in the weak* topology to some measure $\mu \in \mathcal{M}^\sigma(\{0,1\}^{\mathbb{N}})$ that is not ergodic.*

Solution. For each $N \in \mathbb{N}$, consider the following point:

$$x^{(N)} = 0^N \parallel 1^N \parallel 0^N \parallel 1^N \parallel \ldots.$$

Note that the period of this point is exactly $2N$. This allows us to define the following ergodic measures:

$$\mu_N := \frac{1}{2N} \sum_{n=0}^{2N-1} \delta_{\sigma^n(x^{(N)})}.$$

Claim. $\mu_N \xrightarrow[N\to\infty]{} \mu$ where $\mu = \frac{1}{2}\delta_{\mathbf{0}} + \frac{1}{2}\delta_{\mathbf{1}}$ (where $\mathbf{0} = 0000\ldots$ and $\mathbf{1} = 1111\ldots$).

Proof. Let $z \in \{0,1\}^\ell$ be an arbitrary finite binary string of length ℓ and by $\mathsf{C}(z)$ denote the corresponding cylinder set. By definition, $\mu_N(\mathsf{C}(z))$ measures the frequency of the string z appearing as a subblock in the cyclic word $0^N \parallel 1^N$. Without loss of generality, further assume $N > \ell$ (as we are taking a limit, this is legit). Consider three cases.

Case #1. $z = 0^\ell$. We have $\mu_N(\mathsf{C}(0^\ell)) = \frac{N-\ell+1}{2N} \xrightarrow[N\to\infty]{} 1/2$.

Case #2. $z = 1^\ell$. Similarly $\mu_N(\mathsf{C}(1^\ell)) = \frac{N-\ell+1}{2N} \xrightarrow[N\to\infty]{} 1/2$.

Case #3. Since $N > \ell$, transition from 0s to 1s happens exactly once per $2N$ blocks, and the transition from 1s to 0s happens exactly once. Thus, any specific string z other than 0^ℓ or 1^ℓ can appear at most exactly 1 time: $\mu_N(\mathsf{C}(z)) \le \frac{1}{2N} \xrightarrow[N\to\infty]{} 0$.

Note that these three cases produce identical measures to $\mu = \frac{1}{2}\delta_{\mathbf{0}} + \frac{1}{2}\delta_{\mathbf{1}}$, concluding the proof.

Finally note that μ is not ergodic: for instance, the set $\{\mathbf{0}\}$ is σ-invariant but its measure is $\mu(\{\mathbf{0}\}) = 1/2$.

Problem (Exercise 5)**.** *Describe the ergodic measures for the following measure-preserving systems:*

(a) $(\mathbb{R}/\mathbb{Z}, \mathcal{B}, \lambda, x \mapsto x + \alpha)$ *where* $\alpha \in \mathbb{Q}$*;*

(b) The finite space $X = \{1,2,3\}$ *equipped with the normalized counting measure* μ *and the map* T *specified by* $T(1) = 2$, $T(2) = 1$, *and* $T(3) = 3$*;*

(c) $(\mathbb{R}^2/\mathbb{Z}^2, \mathcal{B}, \lambda, (x,y) \mapsto (x + \alpha, -y))$ *where* $\alpha \in \mathbb{R} \setminus \mathbb{Q}$.

Solution.

Part (a). Let $\alpha = p/q$ for $p \in \mathbb{Z}$, $q \in \mathbb{N}$. Without loss of generality, assume $(p,q) = 1$. Under this transformation, every point $x \in \mathbb{R}/\mathbb{Z}$ is strictly periodic with period q. Therefore, we have an uncountable family of ergodic measures parameterized by the base points $x \in [0, 1/q)$:

$$\mu_x = \frac{1}{q}\sum_{n=0}^{q-1} \delta_{x+np/q}.$$

Part (b). Transformation T partitions X into two distinct orbits: one is $1 \xrightarrow{T} 2 \xrightarrow{T} 1$ and the other is $3 \xrightarrow{T} 3$.

Thus, there are exactly two ergodic measures: $\mu_1 = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_2$ and $\mu_2 = \delta_3$.

Part (c). Note that along the first coordinate, the map $x \mapsto x + \alpha$ is uniquely ergodic, while along the second coordinate, values y_0 simply oscilate under T: $y_0 \to -y_0 \to y_0$. Therefore, any ergodic measure in our system must be uniformly spread across the first coordinate and uniformly supported on the given $\{y_0, -y_0\}$. Consequently, we get two cases:

Case #1. $y_0 \in (0, 1/2)$. We have ergodic measures

$$\mu_{y_0} = \lambda \times \left(\frac{1}{2}\delta_{y_0} + \frac{1}{2}\delta_{-y_0}\right).$$

Case #2. $y_0 \in \{0, 1/2\}$. Then, $\mu_{y_0} = \lambda \times \delta_{y_0}$.

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